Mini-Project 1: Image Classification on Caltech-101:

Comparing Classical ML and Deep Learning Approaches

Cindy Sun

SHBT261 — Artificial Intelligence in Medicine

Instructors: Mengyu Wang and Tobias Elze

Harvard University | Spring 2026

GitHub: https://github.com/CindySun0309/SHBT261_miniproject1/tree/main

1. Introduction

Image classification is a core task in computer vision and medical imaging, requiring models to assign categorical labels based on visual content. This project evaluates four classification approaches on the Caltech-101 dataset: a classical machine learning method—Support Vector Machine (SVM) with Histogram of Oriented Gradients (HOG) features—and three pre-trained deep learning models: ResNet-18, EfficientNet-B0, and Vision Transformer Small (ViT-Small).

These models represent three major paradigms in computer vision: hand-crafted features with classical classifiers, convolutional neural networks that learn hierarchical representations directly from data, and Vision Transformers that capture global relationships through self-attention over image patches.

Comparing these approaches is particularly relevant for medical AI, where datasets are often small, interpretability is important, and computational resources may be limited. A systematic comparison was conducted across 107 configurations (4 SVM and 103 deep learning models), evaluating accuracy, generalization, and efficiency to inform practical model selection.

2. Methods

2.1 Dataset

The Caltech-101 dataset from Kaggle consists of 9,146 color images distributed across 101 object categories plus a background class. The background category was excluded because it was not relevant to the study objective, reducing the dataset to 8,677 images. The number of images included in the remaining categories ranges from 40 to 800. This class imbalance is a notable challenge. Images vary substantially in size, viewpoint, illumination, and background complexity.

The dataset was then divided into training (70%, 6,073 images), validation (15%, 1,302 images), and test (15%, 1,302 images) subsets using stratified sampling to preserve class proportions across splits. A fixed random seed (42) was applied throughout to ensure reproducibility. The test set was strictly held out during model development and evaluated only once after training was fully completed.

2.2 Models

SVM + HOG (Baseline)

The classical baseline uses Histogram of Oriented Gradients (HOG) features extracted from grayscale images with 9 orientation bins, 8×8 pixel cells, and 2×2 block normalization, followed by a Support Vector Machine with an RBF kernel (C=10, gamma=scale). Features were standardized using zero-mean unit-variance scaling. For augmented configurations, horizontally flipped copies of training images were appended to the training set. Two image resolutions were tested: 64×64 and 128×128.

ResNet-18

ResNet-18 is a convolutional neural network with 18 layers (~11M parameters) using residual skip connections to enable effective deep feature learning. The model was initialized with ImageNet-pretrained weights. The final fully connected layer was replaced with a dropout layer (p=0.3) followed by a linear layer mapping to 101 classes.

EfficientNet-B0

EfficientNet-B0 (~5.3M parameters) uses compound scaling to balance network depth, width, and resolution. It was similarly initialized with ImageNet weights, with the classifier head replaced by a dropout (p=0.3) and linear output layer.

Vision Transformer Small (ViT-Small/16)

ViT-Small/16 (~22M parameters) divides each 224×224 image into 16×16 patches, linearly embeds them, and processes the resulting sequence with multi-head self-attention. Pre-trained weights from ImageNet-21k were used via the timm library. The classification head was replaced with LayerNorm →

Dropout (p=0.3) → Linear(num_features, 101). ViT requires a fixed 224×224 input and was excluded from the 128×128 grid.

2.3 Training Strategy

All deep learning models were trained using a two-phase transfer learning approach. In Phase 1, the pretrained backbone was frozen and only the classification head was trained. In Phase 2, the entire network was unfrozen and fine-tuned at one-tenth the Phase 1 learning rate. Both phases used the same number of epochs as specified in the hyperparameter grid, with cosine annealing learning rate scheduling applied independently within each phase. Cross-entropy loss with label smoothing (ε=0.1) was used throughout to improve generalization.

2.4 Grid Search Configuration

An exhaustive grid search was performed across the following hyperparameter space:

Parameter	**Values**
Models	ResNet-18, EfficientNet-B0, ViT-Small
Image size	128×128, 224×224 (ViT: 224 only)
Epochs (per phase)	5, 10
Learning rate	1e-3, 1e-4
Optimizer	Adam, AdamW, SGD
Augmentation	True, False
Batch size	32 (fixed)

Table 1. Deep learning hyperparameter grid. A total of 103 DL + 4 SVM = 107 configurations were evaluated.

The grid search was implemented to be resume-safe, with results saved to disk after each run so that experiments could be interrupted and continued without re-training completed configurations.

2.5 Evaluation Metrics

Each model configuration was evaluated on the held-out test set using: overall accuracy, top-5 accuracy, macro-averaged and weighted precision/recall/F1-score, per-class accuracy, and confusion matrix. Training wall-clock time was also recorded to enable efficiency comparisons. The best configuration per model architecture was selected based on maximum test accuracy across all grid search runs.

2.6 Ablation Studies

Four ablation experiments were conducted across all deep learning models to isolate the contribution of individual design choices: (1) image resolution (128×128 vs. 224×224), (2) data augmentation (random horizontal flip, rotation ±10°, color jitter vs. no augmentation), (3) optimizer choice (Adam vs. AdamW vs. SGD with momentum and Nesterov acceleration), and (4) initial learning rate (1e-3 vs. 1e-4). Each ablation averages accuracy across all other hyperparameter dimensions, isolating the effect of the variable of interest.

3. Results

3.1 Best Configuration Per Model

Table 2 summarizes the best-performing configuration per model architecture. All three deep learning models substantially outperformed the SVM baseline, with ViT-Small achieving the highest overall accuracy at 98.23%.

Model	**Accuracy**	**Top-5 Accuracy**	**Marco F1 Score**	**Best Configuration**	**Train Time (minute)**
SVM	67.36%	84.18%	0.531	img=64, aug=False, C=10	3.8
ResNet-18	97.0%	99.77%	0.954	img=224, epochs=5, 1r=0.001, opt=AdamW, aug=False	1.3
EfficientNet-B0	97.39%	99.85%	0.962	img=224, epochs=10, lr=0.001, opt=Adam, aug=True	4.5
ViT-Small	98.23%	99.69%	0.974	img=224, epochs=10, lr=0.001, opt=AdamW, aug=False	7.5

Table 2. Best test-set performance per model. Configurations selected by maximum test accuracy across all grid search runs.

3.2 Overall Comparison

Figure 1 presents three summary panels: (left) best test accuracy per model, (center) accuracy versus training time for all 107 runs, and (right) the effect of data augmentation averaged across hyperparameters. The deep learning models cluster in a narrow 2–8 minute training window while achieving 97–98% test accuracy, whereas the SVM achieves only 67.4% accuracy with longer training time. This observation reveals that deep learning models attain near-peak performance even with short training times. After applying data augmentation, the model's test accuracy decreased slightly compared to training without augmentation, although the overall difference in accuracy was minimal.

Figure 1. (Left) Best test accuracy per model. (Center) Test accuracy vs. training time for all 107 runs. (Right) Average accuracy with and without data augmentation per model.

3.3 Per-Class Accuracy

For visual clarity, Figure 2 displays the top five and bottom five classes by accuracy for each model's best configuration (see Appendix for the full accuracy ranking across all 101 classes). For the SVM, the top classes (laptop, car_side, Faces_easy, scissors, Motorbikes) all achieve perfect or near-perfect accuracy, while the bottom 5 classes (lobster, strawberry, cannon, wild_cat, beaver) score 0%. The deep learning models exhibit a much tighter range — their bottom 5 classes still achieve 62–80% accuracy, confirming far more uniform class-level performance. Classes involving fine-grained aquatic or similar-looking objects (schooner, lobster, anchor, crocodile) appear as the hardest across multiple models.

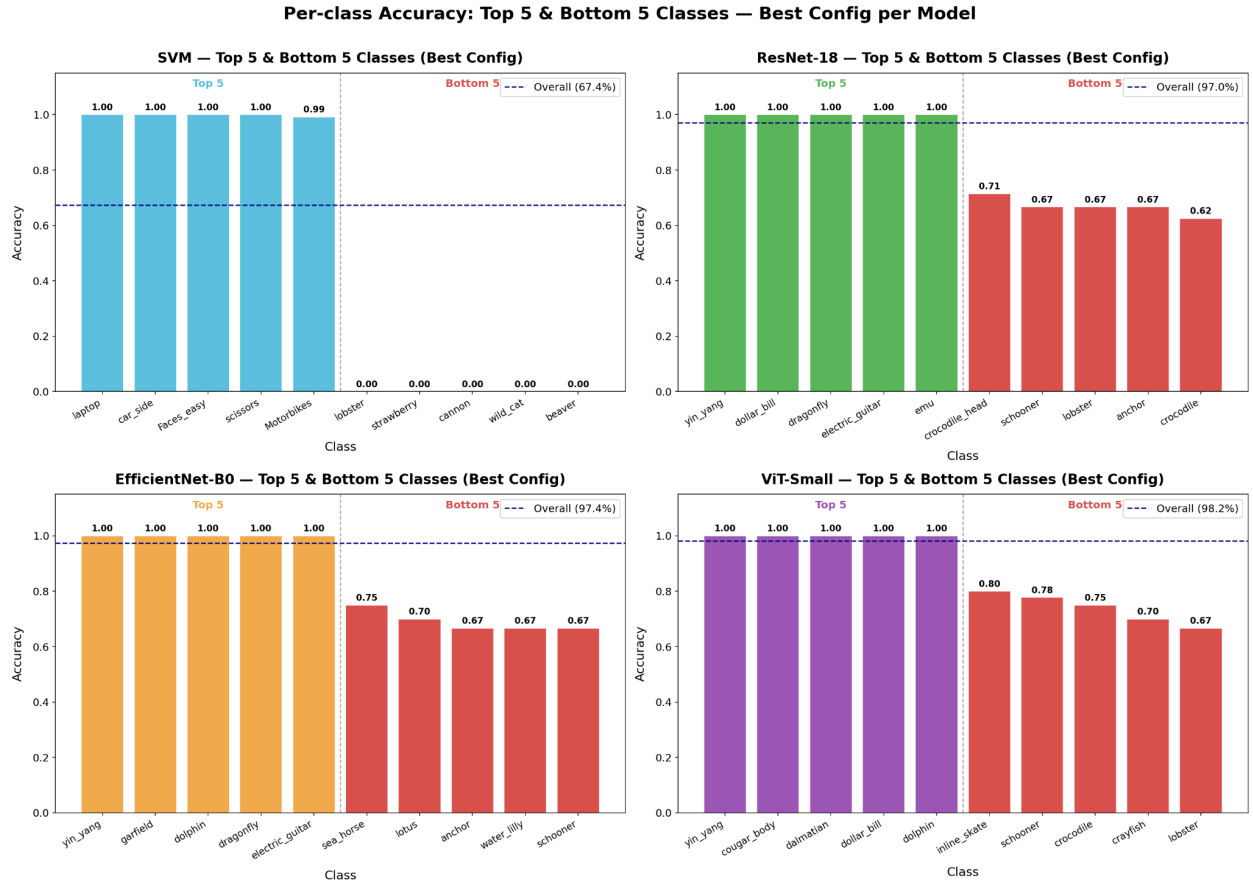
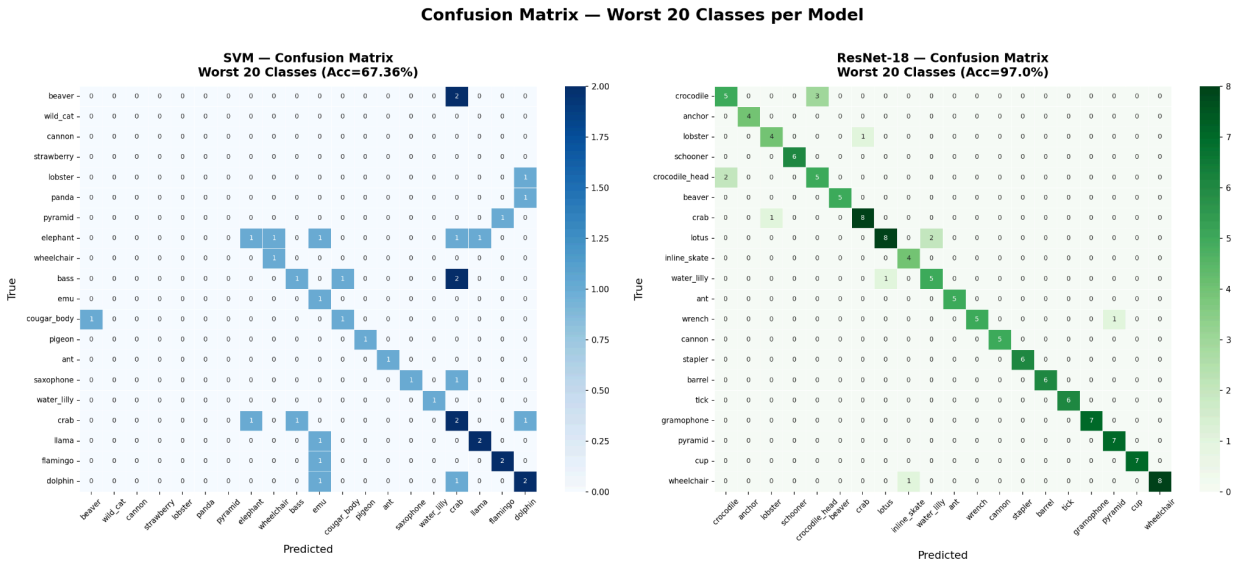


Figure 2. Top 5 and bottom 5 classes by per-class accuracy for each model's best configuration. Blue/green/orange/purple bars indicate top 5; red bars indicate bottom 5. The dashed line shows overall accuracy.

3.4 Confusion Matrix

Due to the 101-class scale, a reduced 20×20 confusion matrix restricted to the worst-performing classes is shown in Figure 3, with cell counts annotated (full 101×101 matrices and top-15 misclassified pair plots are provided in the Appendix). For the SVM, errors are spread across many classes with no dominant confusion cluster. For the deep learning models, errors are sparse and concentrated in a small number of structurally similar class pairs, confirming that remaining misclassifications are semantically reasonable.



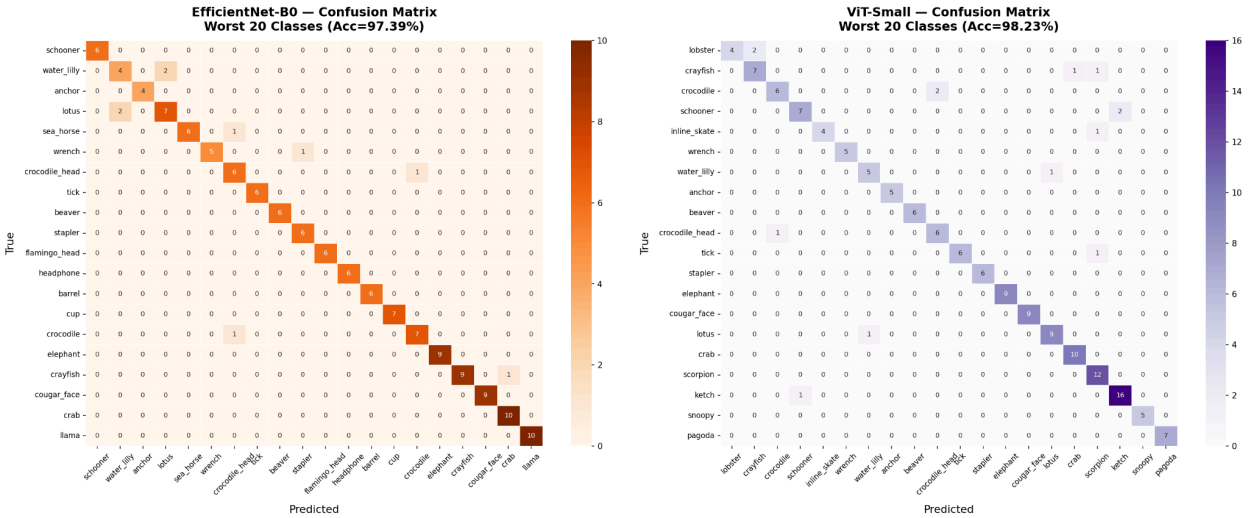


Figure 3. Confusion matrix restricted to the 20 worst-performing classes per model, with cell counts annotated. Rows = true class, columns = predicted class.

3.5 Ablation Studies

Figure 4 presents the four ablation panels, each averaging accuracy across all other hyperparameter dimensions.

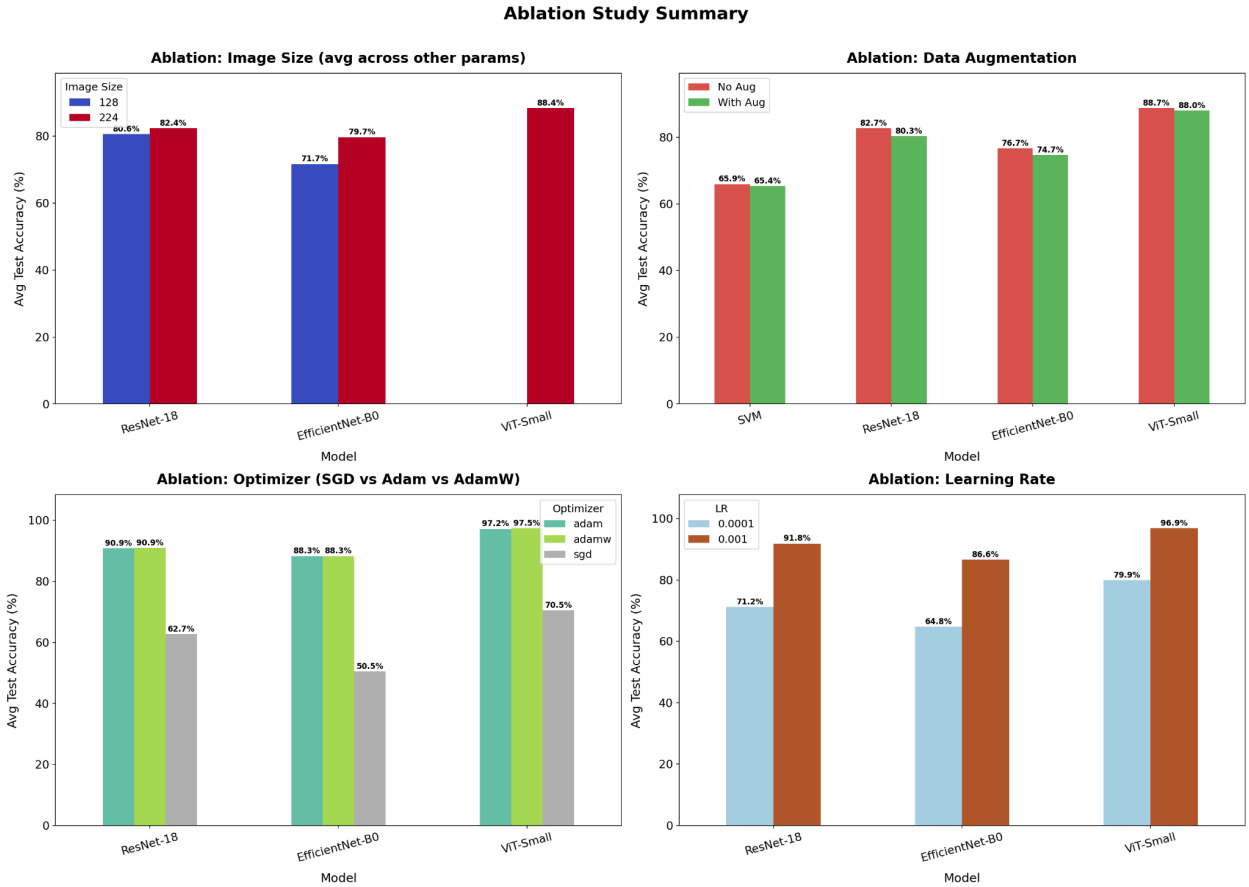


Figure 4. Ablation study summary. (Top-left) Image resolution effect. (Top-right) Data augmentation effect. (Bottom-left) Optimizer comparison. (Bottom-right) Learning rate comparison.

Image Resolution: 224×224 consistently outperforms 128×128 across all deep learning models. The effect is most pronounced for EfficientNet-B0, which gains approximately 9 percentage points on average (71% → 80%). ResNet-18 shows a smaller gain (~1%), suggesting its architecture is more resolution-robust due to global average pooling.

Data Augmentation: Augmentation had a neutral to slightly negative effect across all models and the SVM. With augmentation, EfficientNet-B0 drops ~2%, ResNet-18 drops ~2%, and ViT-Small is nearly unchanged. This likely reflects the characteristics of Caltech-101: images are relatively clean and centered, and the dataset is small enough that augmentation-induced variance may exceed any regularization benefit.

Optimizer: Adam and AdamW substantially outperform SGD across all models. The gap is largest for EfficientNet-B0, where SGD averages only ~50% compared to ~88% for Adam and AdamW. SGD without a warm-up schedule struggles to fine-tune pretrained weights effectively, as its uniform learning rate and momentum are poorly suited to the non-convex loss landscape of transfer learning.

Learning Rate: An initial learning rate of 1e-3 dominates 1e-4 by a large margin for all three models (~20–25 percentage point gap for ViT-Small and ResNet-18). The two-phase training strategy — which applies a 10× reduction in Phase 2 — compensates, making the higher initial rate safe and beneficial for rapid head convergence in Phase 1.

4. Discussion

4.1 Classical vs. Deep Learning

The 30+ percentage point accuracy gap between SVM+HOG (67.4%) and the deep learning models (97–98%) confirms the well-established superiority of learned representations for multi-class object recognition. HOG captures local edge orientation statistics but cannot encode semantic or global structure, which is essential for discriminating 101 diverse object categories. The per-class accuracy analysis reveals that SVM struggles specifically on fine-grained or texture-heavy classes — precisely those for which local gradient histograms provide insufficient discriminative signal. The confusion matrix further reveals that SVM errors are sparse and scattered across many class pairs with no dominant confusion cluster, whereas DL model errors are concentrated in a small number of semantically related pairs (e.g., schooner↔ketch, crocodile↔crocodile_head, lobster↔crayfish), with most off-diagonal cells remaining at zero.

4.2 CNN vs. Vision Transformer

ViT-Small achieved the highest accuracy (98.23%, Macro F1 = 0.974), marginally outperforming both CNNs. This finding aligns with the general trend that Vision Transformers, when pretrained on large-scale datasets (ImageNet-21k in this case), can outperform CNNs on downstream tasks. The attention mechanism enables ViT to model long-range spatial dependencies from the first layer, which may be advantageous for object categories where global configuration is more diagnostic than local texture. However, ResNet-18 achieves 97.0% accuracy with a training time of only 1.3 minutes — roughly 6× faster than ViT-Small. Given the marginal accuracy difference, ResNet-18 represents an outstanding efficiency-accuracy tradeoff and would be the recommended choice for resource-constrained deployment. EfficientNet-B0 occupies an intermediate position, achieving 97.4% in 4.5 minutes with a smaller parameter count (~5.3M vs. ~11M for ResNet-18).

4.3 Implications for Medical AI

The results carry several practical implications for medical image classification. First, the effectiveness of pretrained models even with brief fine-tuning (5–10 epochs per phase) suggests that transfer learning

from large natural image datasets can be highly effective even when labeled medical data is scarce. Second, the failure of data augmentation to improve performance on Caltech-101 warrants caution: in medical imaging, domain-appropriate augmentation may behave differently from the generic transformations applied here. Third, the strong sensitivity to optimizer choice (Adam/AdamW >> SGD) reinforces best practices for transfer learning in the medical imaging literature.

4.4 Limitations

Several limitations should be noted. Caltech-101 is relatively clean and well-centered, unlike many medical imaging datasets that contain acquisition artifacts, noise, and significant intra-class variation. Additionally, only 5 and 10 training epochs per phase were tested; longer training may further improve performance, particularly for ViT. The grid search fixed batch size at 32 and weight decay at 1e-4 without ablation, which may have constrained certain configurations.

5. Conclusion

This project conducted a systematic comparison of four image classification approaches across 107 hyperparameter configurations on the Caltech-101 benchmark. The key findings are:

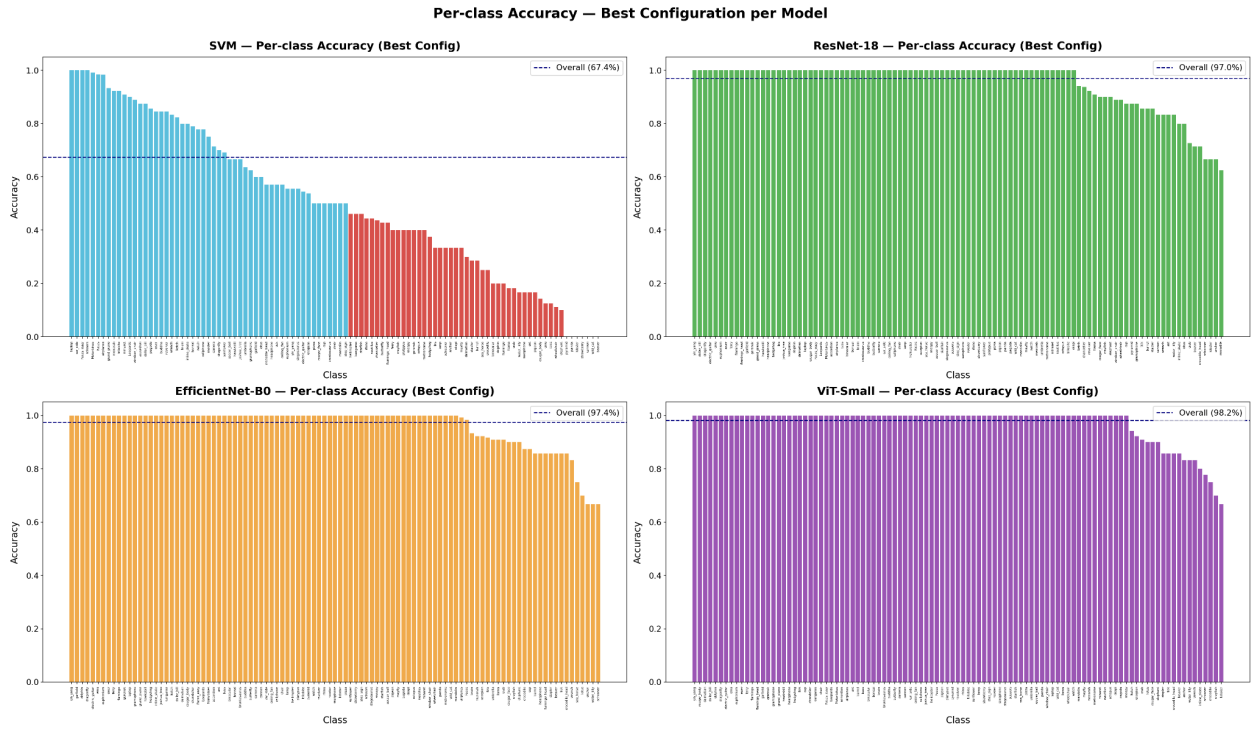
- Deep learning models dramatically outperform the classical SVM+HOG baseline (97–98% vs. 67.4%), confirming the necessity of learned representations for complex multi-class recognition.
- ViT-Small achieves the best accuracy (98.23%), but ResNet-18 offers the best efficiency-accuracy tradeoff (97.0% in 1.3 minutes), making it the recommended choice for resource-constrained settings.
- Data augmentation did not improve performance on this dataset, suggesting its benefit is dataset- and domain-dependent.
- Adam and AdamW consistently outperform SGD for transfer learning, and a higher initial learning rate (1e-3) is preferable within the two-phase training framework.
- Misclassification analysis reveals that remaining DL errors are semantically reasonable (e.g., schooner↔ketch), whereas SVM errors are more arbitrary — a meaningful distinction for safety-critical applications.

These findings inform principled model selection for medical AI: transfer learning from large pretrained models is highly effective even with limited data and short training, optimizer and resolution choices matter substantially more than augmentation, and targeted misclassification visualization (confusion pairs + worst-class zoom) is more actionable than a full confusion matrix for high-class-count problems.

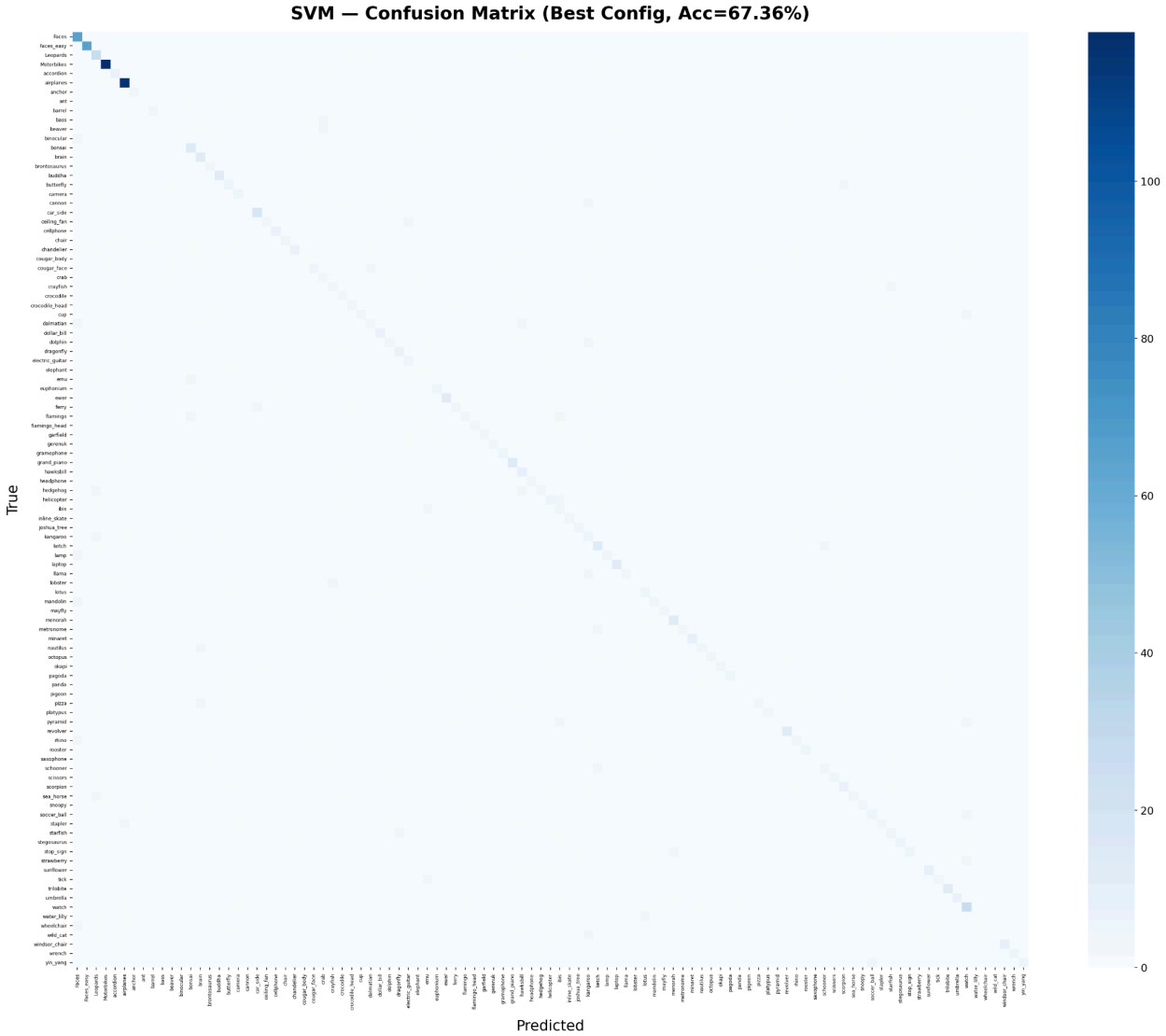
Appendix

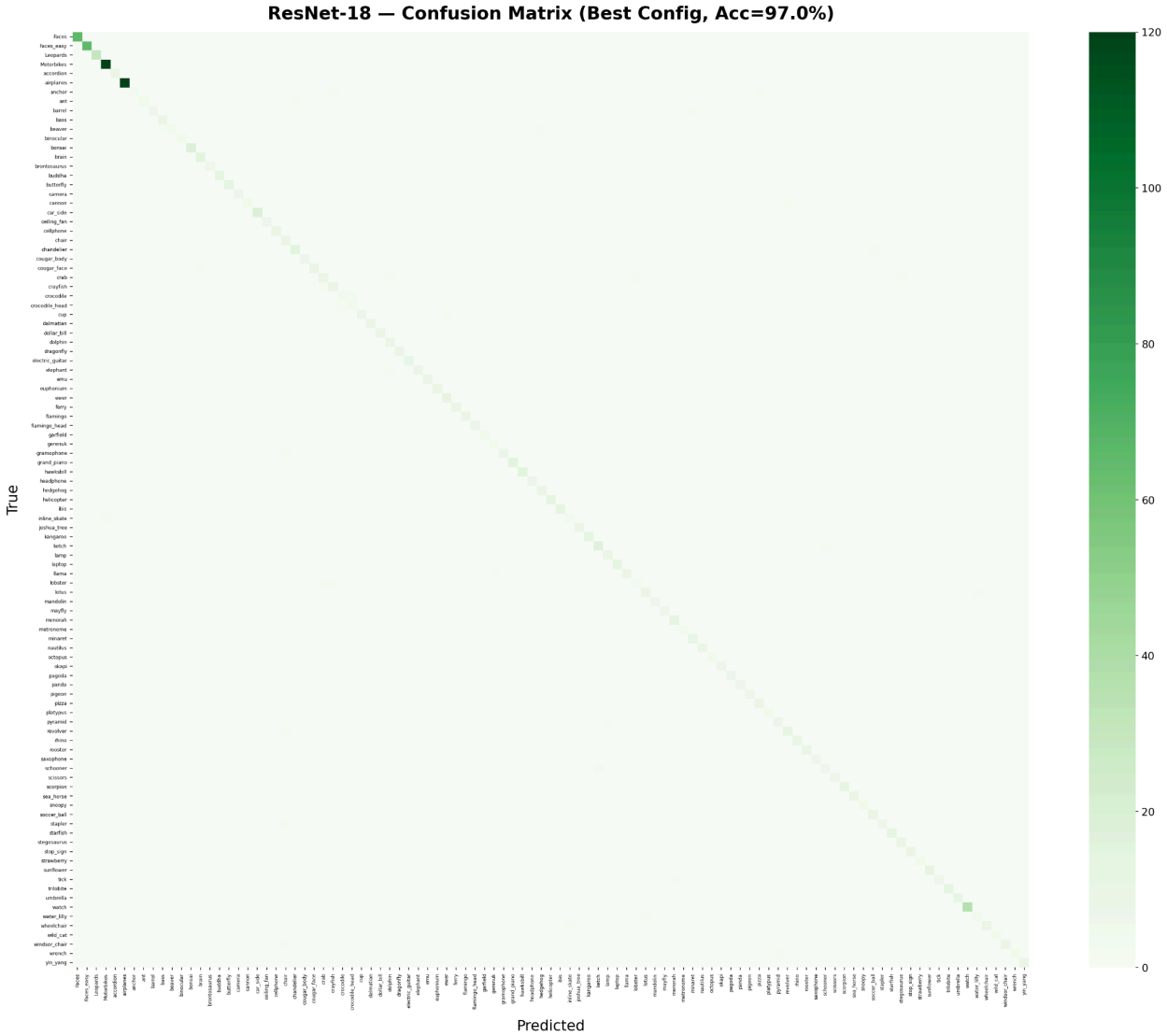
See below for complementary plots and tables that were omitted from the main report for clarity and space limitations.

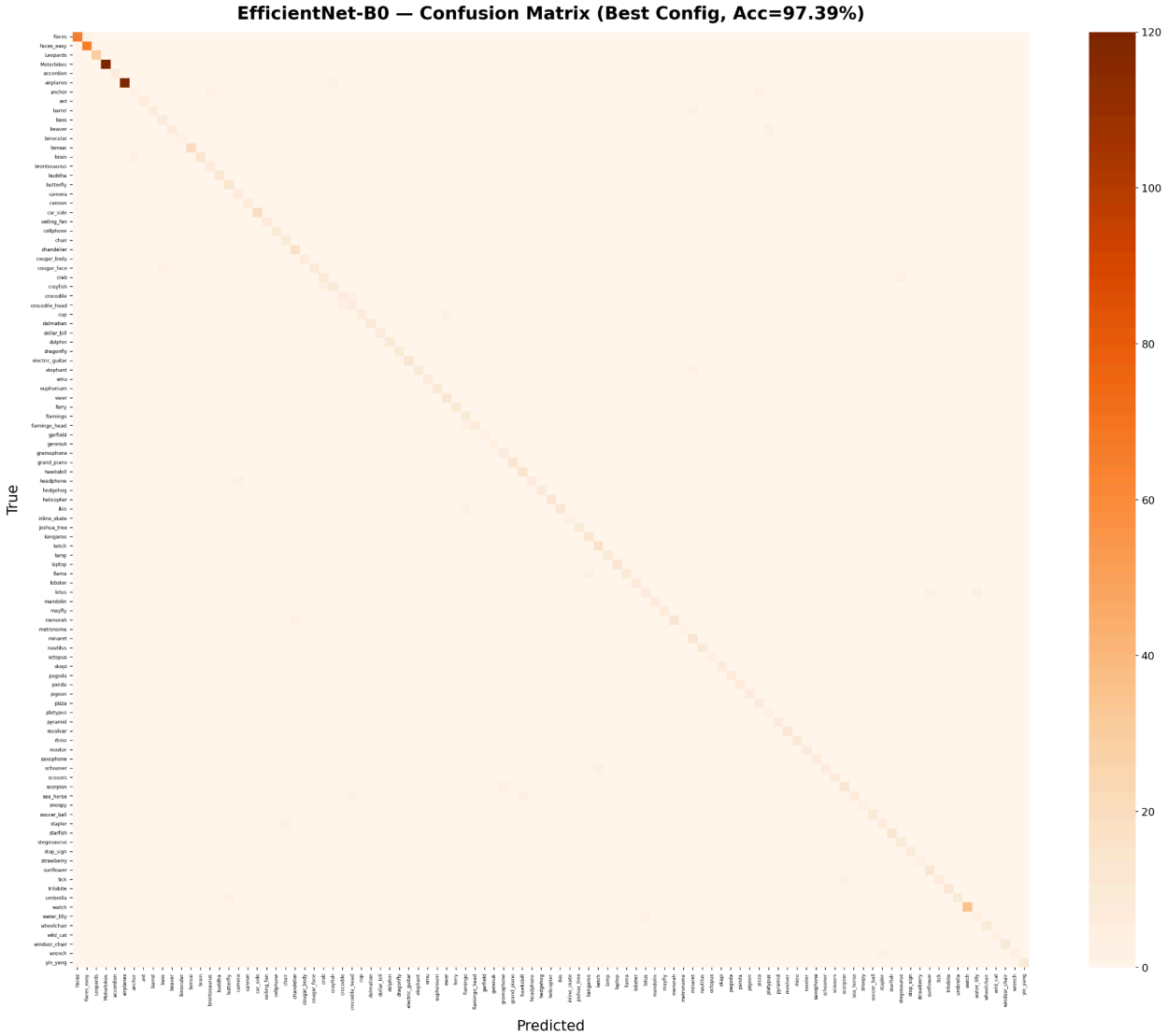
Full Accuracy Ranking Across All 101 Classes (3.3 Per-Class Accuracy)

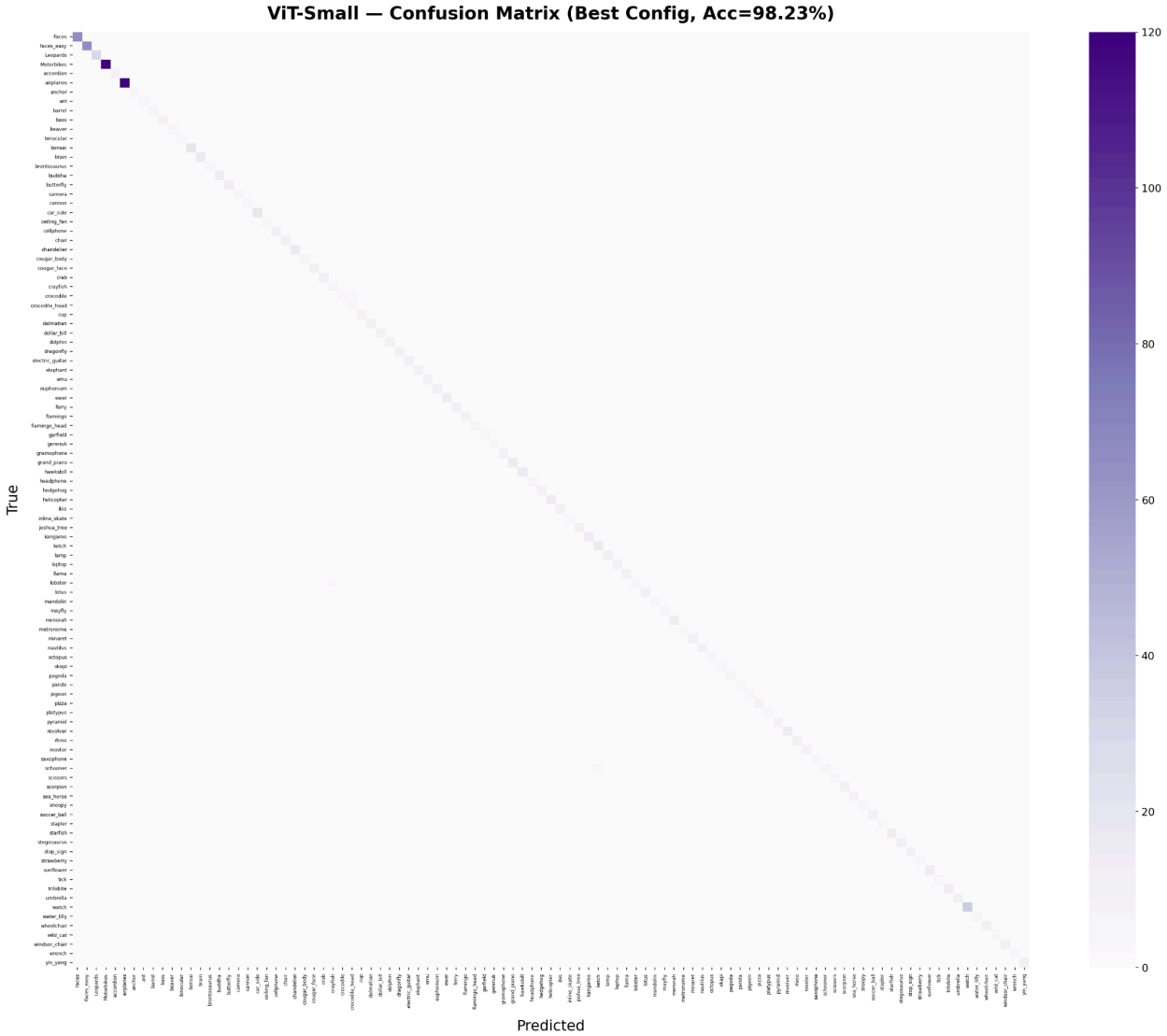
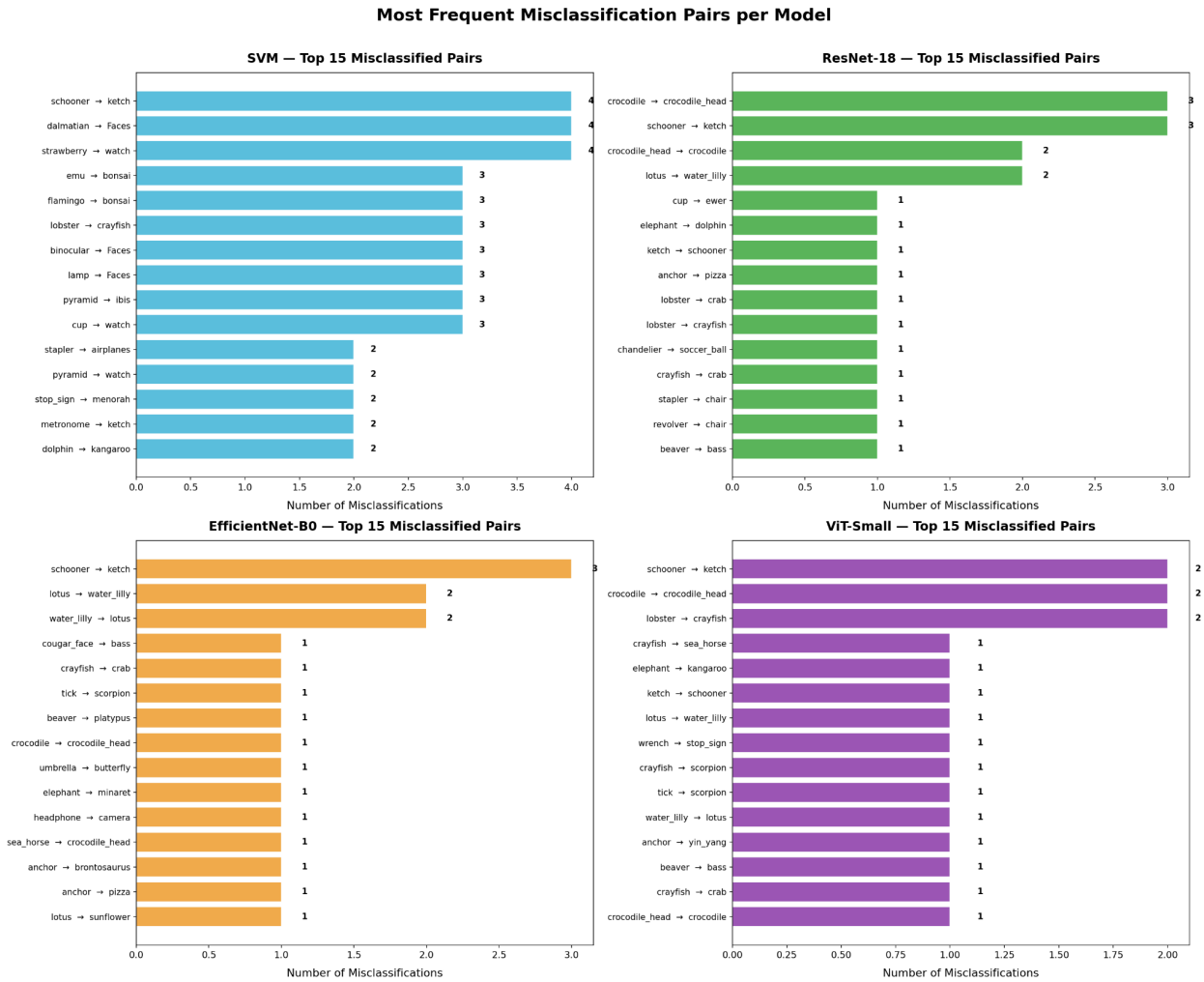


Full 101×101 Matrices (3.4 Confusion Matrix)




ResNet-18 — Confusion Matrix (Best Config, Acc=97.0%)
True
Predicted
120
100
80
60
40
20
0


EfficientNet-B0 — Confusion Matrix (Best Config, Acc=97.39%)
True
Predicted
120
100
80
60
40
20
0


ViT-Small — Confusion Matrix (Best Config, Acc=98.23%)
True
Predicted
120
100
80
60
40
20
0

Top-15 Misclassified Pair Plots (3.4 Confusion Matrix): Top 15 most frequent misclassification pairs per model (true class → predicted class). Diagonal entries are excluded to focus on errors only.